∴ G1

A	B	C	D	E	F	G
15	3	8	17	∞	11	

∴ total = ∞

∴ G2

A	C	E	G	B	D	F
14	3	11	∞	9	∞	

∴ total = ∞

∴ G3

B	E	A	F	D	G	C
6	∞	∞	17	9	2	

∴ total = ∞

∴ G4

C	A	G	F	D	E	B
14	∞	10	∞	∞	8	

∴ total = ∞

$\boxed{\text{2 fittest} = G1 \ \& \ G2}$ (Ans)

3|

G1

A	B	C	D	E	F	G

G2

A	C	E	G	B	D	F

∴ off-1

A	B	C	G	E	D	F

off-2

A	C	E	D	E	F	G

No, they are not eligible because, the Offsprings contain duplicate visits. Only 1 time visit is allowed

4|

No, in usual mutation, we only change bit of chromozome randomly. But in this problem, we can't, as duplicate visits are not allowed.

3||

	A	B	C	D	E	F	G	H
C_2	0	1	1	0	0	1	0	1
C_3	1	0	0	0	1	0	0	1

m-point

	A	B	C	D	E	F	G	H
∴ off-1	0	1	1	0	1	0	0	1
∴ off-2	1	0	0	0	0	1	0	1

m-point

4||

Mutant:

	A	B	C	D	E	F	G	H
off-1	0	1	1	1	1	0	0	1
off-2	1	0	0	0	1	1	0	1

total = (2+3+8+7+2) = 22 [rejected]

total = (1+7+4+2) = 14 [rejected]

off-1, D point, and off-2 E point bit was flipped. Then weight was calculated, as they were more than 12, both offsprings got rejected.

B||

1.

G1	A	B	C	D	E	F	G
G2	A	C	E	G	B	D	F
G3	B	E	A	F	D	G	C
G4	C	A	G	F	D	E	B

2.

fitness = ~~Σ short~~

min (Σ shortest dista between 2 cities)

Practice Problems

A||

1. C_1

A	B	C	D	E	F	G	H
0	1	0	1	0	1	0	1

C_2

A	B	C	D	E	F	G	H
0	1	1	0	0	1	0	1

C_3

A	B	C	D	E	F	G	H
1	0	0	0	1	0	0	1

C_4

A	B	C	D	E	F	G	H
1	1	1	1	1	1	1	1

$\therefore C_2$

A	B	C	D	E	F	G	H
1	0	0	0	1	0	0	1
20				15			7

$\therefore$ Total $= 1+7+2$
$= 10$

$\therefore$ Fitness $= (20+15+7)$
$= 42$

$\therefore C_4$

A	B	C	D	E	F	G	H
1	1	1	1	1	1	1	1

$\therefore$ Total $= (1+2+3+8+7+4+5+2)$
$= 32$ [rejected]

2|| fitness = if (Σweight) ≤ 12; then max(Σreward)

$\therefore C_1$

A	B	C	D	E	F	G	H
0	1	0	1	0	1	0	1

total $= (2 + 8 + 4 + 2)$
$= 16$ [rejected]

$\therefore C_2$

A	B	C	D	E	F	G	H
0	1	1	0	0	1	0	1

total $= (2+8+4+2)$
~~$= 18$ [rejected]~~
$= 11$

$\therefore$ fitness $= (8+10+25+7)$
$= 47$

$\therefore$ ~~fitness~~

$\therefore$ Fittest 2:

C_2, C_3 (Ans)

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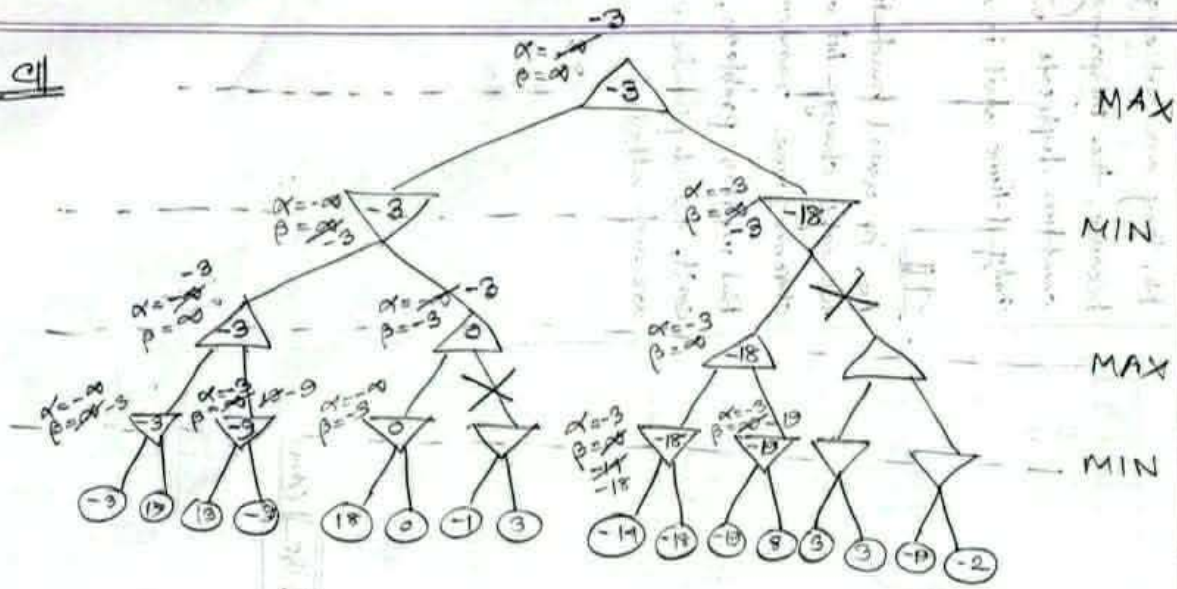
$\alpha \geq \beta$

Q1



$23 \leq (10 + 13)$ $\quad 12+12$

$D = 22$, $h(B) \leq h(B,H) + h(H)$ $\quad C = 25 \leq 25 \leq 12$

$h(B,G) + h(G)$ $\quad (14+11) = 25$

$15 + 11$

C||

2||

A	45	F	12
B	23	G	11
C	25	H	13
D	22	I	0
E	17		

∴ The values are admissible. (ans)

1||

A	45	D	22	G	11
B	23	E	17	H	13
C	25	F	12	I	0

∴ The values are consistent. (Ans)

3||

A	45	D	22	G	5
B	23	E	17	H	13
C	25	F	12	I	0

For Node C, $h(C) \leq h(C,G) + h(G)$

$\Rightarrow 25 \leq (14+5) = 19$

$\Rightarrow 25 > 19$ [∴ not consistant]

∴ The values are admissible but not consistent. (Ans)

B)

A^* search

S	7
A	9
B	4
C	2
D	5
E	3
G	0

	~~S 0+7~~
S^7	$A^{3+9=12}$ ~~$D^{2+5=7}$~~
D^7	A^{12} ~~$B^{7+3=7}$~~ $E^{3+6=9}$
B^7	A^{12} E^9 ~~$C^{2+5=7}$~~ $E^{3+4=7}$
C^7	A^{12} E^9 ~~E^7~~ G^{0+9}
E^7	A^{12} E^9 G^9 ~~$G^{0+7=7}$~~
G^7	A^{12} E^9 G^9

∴ A^* path:

$S \xrightarrow{2} D \xrightarrow{1} B \xrightarrow{1} E \xrightarrow{3} G$

∴ $\boxed{\text{Cost} = 2+1+1+3 = 7}$ (Ans)

GBFS

	~~S^7~~
S^7	A^9 ~~D^5~~
D^5	A^9 B^4 ~~E^3~~
E^3	A^9 B^4 ~~G^7~~
G^0	A^9 B^4

∴ GBFS path:

$S \xrightarrow{2} D \xrightarrow{1} E \xrightarrow{3} G$

∴ $\boxed{\text{Cost} = 2+1+3 = 9}$ (Ans)

A1

Assignment-1

A* search

A	10
B	8
C	5
D	7
E	3
F	6
G	5
H	3
I	1
J	0

$\boxed{A^{0+10}}$

$\Rightarrow A^{10}$ $\boxed{B^{8+6=14} \quad \cancel{F^{6+3=9}}}$

$\Rightarrow F^{9}$ $\boxed{B^{14} \quad \cancel{G^{5+4=9}} \quad H^{3+10=13}}$

$\Rightarrow G^{9}$ $\boxed{B^{14} \quad H^{13} \quad \cancel{I^{1+7=8}}}$

$\Rightarrow I^{8}$ $\boxed{B^{14} \quad H^{13} \quad E^{3+12=15} \quad \cancel{J^{0+10=10}} \quad H^{3+9=12}}$

$\Rightarrow J^{10}$ $\boxed{B^{14} \quad H^{13} \quad E^{15} \quad H^{12}}$

$\therefore$ A* path $= A \xrightarrow{3} F \xrightarrow{1} G \xrightarrow{3} I \xrightarrow{3} J$

$\therefore$ $\boxed{\text{cost} = 10}$ (Ans)

GBFS

$\boxed{\cancel{A^{10}}}$

A^{10} $\boxed{B^{8} \quad \cancel{F^{6}}}$

F^{6} $\boxed{G^{5} \quad \cancel{H^{3}}}$

H^{3} $\boxed{G^{5} \quad \cancel{I^{1}}}$

I^{1} $\boxed{G^{5} \quad E^{3} \quad \cancel{J^{0}}}$

J^{0} $\boxed{G^{5} \quad E^{3}}$

$\therefore$ GBFS path $=$

$A \xrightarrow{3} F \xrightarrow{7} H \xrightarrow{2} I \xrightarrow{3} J$

$\therefore$ $\boxed{\text{cost} = (3+7+2+3) = 15}$ (Ans)

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